

AAYUSH NITIN MAHESHWARKAR

Senior Software Engineer · Java · Spring Boot · Microservices · Distributed Systems · Agentic AI
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PROFESSIONAL SUMMARY

Senior Software Engineer with 5+ years of experience contributing to large-scale, revenue-generating enterprise platforms. Skilled in Java, Python, Spring Boot, and REST APIs, with a strong track record of cross-functional delivery of complex, high-stakes features across global systems. Hands-on experience in Agentic AI instruction design, production hardening, and the full software development lifecycle. AWS Solutions Architect (in progress).

EXPERIENCE

Senior Software Engineer · Federal Express Corporation

Jul 2021 – Present

Orlando, FL · Promotion: Full Stack Developer II → III (Feb 2023)

- **Revenue Impact:** Over \$600M in annualized revenue across enterprise Java and Spring Boot rating platforms handling 2.5M+ peak daily rate requests, spanning pricing, surcharge, and dangerous-goods initiatives.
- **Quote Platform (QP):** Modernized backend rating logic in Java and Spring Boot to support the Quote Platform initiative, enabling one-off spot rate calculations outside of contractual pricing and contributing to \$450M in billed revenue and \$1.8BN in quoted revenue across 69 markets in FY23.
- **Surcharge calculation engine — \$150M+ annualized revenue:** Built core Java/Spring Boot engine exposed via REST APIs for high-throughput surcharge assessment; integrated into the 2.5M req/day rating pipeline.
- **Dangerous Goods surcharge — \$41M direct + \$1.2B linked revenue:** Implemented precedence logic in Java/Spring Boot for the new dangerous-goods surcharge category, governing how it stacked with existing surcharges across country-specific shipping rules; enabled migration of 34,000+ enterprise customers.
- **~600ms latency reduction on customer-facing rate-quote API:** Designed proactive Okta JWT pre-caching using Caffeine refreshAfterWrite with async refresh, eliminating token-fetch overhead from the critical rating path; pattern adopted by adjacent teams.
- **Production AI governance — Microsoft Copilot:** Owned quality standards and final-approval authority for LLM instruction sets across multiple enterprise platforms; defined prompt evaluation criteria for clarity, completeness, security, and accuracy, gating git commits on benchmark adherence.
- **Agentic AI pipeline — LangChain + Claude:** Contributed to a LangChain and Claude-powered agentic pipeline built to handle a projected 10x surge in error-correction volume from 2,000 to 20,000 daily shipments, eliminating a 5-day manual remediation cycle. Implemented configurable database-level retry controls and authored Python automation for bulk data rectification.
- **Cache optimization across REBS services:** Implemented Caffeine Loading Cache by identifying ideal cache candidates to maintain pre-loaded state, reducing downstream service calls and improving rating pipeline performance.

Teaching Assistant · Rensselaer Polytechnic Institute

Aug 2019 – Dec 2020

Computer Organization · Intro to Formal Logic and AI · Troy, NY

- Mentored 300+ students across Computer Organization (C, MIPS, x86 assembly) and Intro to Formal Logic and AI; conducted code reviews and supported proof construction in HyperSlate.

TECHNICAL SKILLS

Languages: Java, Python, SQL, Shell Scripting, C

Backend & Microservices: Spring Boot, Spring Cloud, Spring Security, Hibernate/JPA, REST APIs, Feign, Resilience4j (Circuit Breakers, Retry), JMS Messaging, Service Discovery (Eureka), Event-driven Architecture

Cloud & DevOps: AWS Solutions Architect Associate (Certification in Progress), Docker, Jenkins CI/CD, Gradle, Linux, Git

Security: OAuth 2.0, SSO, Okta, JWT, Spring Security

Databases & Caching: PostgreSQL, Oracle, MySQL, MongoDB, Redis, Caffeine (Loading Cache, refreshAfterWrite)

Observability: Splunk, Dynatrace, Grafana, AppDynamics

AI & LLMs: Agentic AI System Design, AI Governance, LangChain, LLM Instruction Engineering, Prompt Evaluation, RAG Fundamentals, GRPO Fine-Tuning Fundamentals, Microsoft Copilot, Claude

ACHIEVEMENTS

- FedEx Hall of Fame Award (FY24): Recognized for contributions to the Quote Platform initiative — \$450M billed / \$1.8B quoted across 69 global markets in FY23.
- AAAI 2020 Fall Symposium on AI for Social Good (Co-Author): "Socioeconomic and Geographic Variations Impacting the Spread of Malaria." [\[GitHub\]](#)

EDUCATION

M.S. Information Technology · Rensselaer Polytechnic Institute · Troy, NY · CGPA 3.8/4.0

Aug 2019 – Dec 2020

B.Tech Electronics & Communication (Minor: CS) · PES University · Bengaluru · CGPA 7.63/10 · Minor CGPA 8.95/10 Aug 2015 – Jul 2019